

DROBETA-TURNU SEVERIN, RomaniaWMO: **154100**Lat: **44.6264N**Lon: **22.6261E**Elev: **257**StdP: **14.56**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.3	18.7	7.2	8.0	19.3	12.0	10.2	22.8	27.0	34.3	23.4	34.0	2.9	10	0.428

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	94.9	69.6	92.0	69.3	89.0	68.5	73.4	86.9	71.8	85.7	70.3	83.9	4.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.2	108.3	79.1	67.5	102.0	77.3	65.9	96.5	75.6	37.1	86.6	35.6	86.3	34.4	84.0	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.2	17.9	14.4	DB	8.4	99.1	5.4	3.8	4.6	101.9	1.4	104.1	-1.6	106.3	-5.6	109.1	(4)
(5)			WB	7.8	75.3	5.3	2.5	4.0	77.1	0.9	78.5	-2.1	79.9	-5.9	81.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.3	33.2	37.3	46.0	55.8	64.7	71.6	75.6	75.8	66.3	55.6	45.1	35.4		
	DBStd	16.50	7.85	8.76	7.78	6.29	5.85	5.75	5.16	5.34	6.51	7.17	7.06	7.45		
	HDD50	1731	521	359	169	20	0	0	0	0	0	30	179	453		
	HDD65	4604	986	775	590	282	80	12	1	2	62	300	597	917		
	CDD50	3664	1	5	43	193	455	648	795	799	490	203	31	1		
	CDD65	1064	0	0	0	5	70	210	331	337	103	8	0	0		
	CDH74	10399	0	0	2	89	638	2040	3395	3321	833	81	0	0		
	CDH80	4341	0	0	0	15	127	798	1575	1550	266	10	0	0		
(14) Wind	WSAvg	4.4	4.2	4.8	5.3	4.7	4.3	4.5	4.6	4.1	4.2	3.9	4.0	4.3		
(15) Precipitation	PrecAvg	27.0	1.8	1.7	1.7	2.4	2.7	2.8	2.7	1.8	2.6	2.2	2.1	2.6		
	PrecMax	46.7	3.5	5.4	4.2	5.4	6.4	7.0	9.8	6.7	9.0	6.0	4.7	7.2		
	PrecMin	13.2	0.2	0.1	0.0	0.1	0.9	0.3	0.3	0.1	0.2	0.0	0.0	0.0		
	PrecStd	6.7	1.0	1.3	1.1	1.2	1.4	1.7	2.0	1.5	1.9	1.6	1.4	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	64.1	72.7	82.4	86.4	95.0	99.8	99.0	91.4	81.3	68.7	56.7		
		MCWB	46.3	50.3	53.6	62.0	65.2	70.3	69.0	70.2	67.2	62.7	56.5	47.6		
	2%	DB	51.3	58.6	67.7	76.2	83.4	91.3	95.1	95.2	86.7	75.8	62.6	51.1		
		MCWB	43.8	47.9	52.1	58.9	64.6	69.8	69.3	69.9	65.7	60.9	53.4	45.1		
	5%	DB	46.8	53.6	63.0	71.8	80.8	88.0	91.9	92.0	82.7	71.2	58.5	47.9		
		MCWB	41.4	45.1	49.7	56.3	63.6	69.0	69.4	69.3	64.2	59.1	52.2	42.9		
	10%	DB	43.3	49.2	58.8	68.0	77.7	84.9	88.7	88.6	78.8	66.8	55.3	45.1		
		MCWB	39.6	42.9	47.8	54.8	62.4	68.0	68.7	68.3	63.1	57.3	50.6	41.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.6	51.6	55.4	63.6	69.5	74.6	75.1	76.7	70.6	65.8	57.5	49.1		
		MCDB	55.4	61.3	68.5	77.5	82.1	88.0	89.1	90.1	84.8	74.1	64.2	54.7		
	2%	WB	44.6	48.7	53.0	60.7	66.9	72.4	72.9	73.5	67.5	62.3	55.4	46.2		
		MCDB	49.9	56.3	64.7	72.3	79.1	86.1	86.8	86.3	80.3	71.7	60.6	50.1		
	5%	WB	42.2	46.0	50.9	58.4	65.2	70.7	71.5	71.4	65.8	60.2	53.3	43.7		
		MCDB	46.0	51.7	61.3	68.4	77.0	84.0	85.5	85.7	78.1	69.2	57.4	47.0		
	10%	WB	39.8	43.8	48.9	56.2	63.5	69.1	70.2	69.8	64.4	58.1	51.2	41.7		
		MCDB	43.2	49.1	57.1	65.3	74.6	81.8	83.7	83.7	75.8	65.5	54.9	44.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	13.7	17.4	18.9	19.6	20.8	21.9	21.6	18.8	16.5	12.1	10.6		
		MCDBR	18.4	22.7	25.9	26.5	25.3	25.7	27.0	26.9	25.1	24.0	19.1	15.5		
		MCWBR	12.0	13.2	13.0	11.8	10.2	9.3	8.2	8.1	8.9	11.0	10.9	10.1		
		MCDBR	16.5	19.8	23.1	22.9	22.6	23.7	23.6	23.9	21.8	21.8	15.9	13.6		
	5% WB	MCWBR	11.1	12.0	12.1	11.0	10.0	9.5	8.3	8.6	8.5	10.4	9.6	9.4		
(40) Clear-Sky Solar Irradiance	taub		0.344	0.375	0.415	0.479	0.459	0.451	0.465	0.485	0.449	0.407	0.375	0.344		
	taud		2.268	2.175	2.152	2.003	2.122	2.192	2.159	2.096	2.157	2.265	2.295	2.302		
	Ebn at Noon		237	250	256	249	259	261	256	245	243	237	223	224		
	Edh at Noon		27	36	42	53	48	45	46	48	41	33	27	24		
(44) All-Sky Solar Radiation	RadAvg		428	679	1066	1452	1770	1990	1981	1759	1247	809	462	363		
	RadStd		60	96	143	161	150	174	136	165	116	110	67	54		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.26	N/A	N/A	N/A	N/A	N/A	N/A	-288	+250	+133
(47) Regional (63 neighbors)		+1.13	N/A	N/A	N/A	N/A	N/A	N/A	-281	+191	+104

Nomenclature: See separate page